

The ReLU Frequency Tradeoff: Why Neural Networks Prefer Low Frequencies

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Abstract

We prove that single-hidden-layer ReLU networks with H hidden units have at most $H+1$ linear pieces and at most H zeros. This directly limits their ability to approximate high-frequency functions: a sine wave $\sin(2\pi\omega x)$ has 2ω zeros, requiring $H \geq 2\omega$ for accurate approximation. We establish three theorems: (1) an exact breakpoint count showing H units produce at most $H+1$ pieces; (2) a zero-count bound showing H units produce at most H zeros; and (3) a frequency barrier showing that under-capacity networks ($H < 2\omega$) produce fundamentally larger approximation errors than sufficient-capacity networks ($H \geq 2\omega$). All theorems are verified via analytical constructions on NVIDIA Jetson Orin GPU — no gradient descent needed.

1. Introduction

ReLU networks are universal approximators — but with what cost? A single-hidden-layer network with H units has only $H+1$ linear pieces (Pascanu et al., 2013). This seemingly simple fact has profound consequences for high-frequency approximation.

Consider the target function $g(x) = \sin(2\pi\omega x)$. It has 2ω zeros and oscillates ω times per unit interval. A ReLU network must match these oscillations with its piecewise linear structure. But if the network has only H pieces, it can have at most H zeros. When $H < 2\omega$, the network simply cannot match the target’s oscillation count — approximation is fundamentally limited.

We prove this limitation and quantify the frequency barrier.

1.1 Contributions

1. **Breakpoint Count (Theorem 1):** A ReLU network with H hidden units has exactly at most H breakpoints and $H+1$ linear pieces. This is an exact combinatorial bound.
2. **Zero Count (Theorem 2):** The same network has at most H zeros (sign changes). Each linear piece can cross zero at most once.
3. **Frequency Barrier (Theorem 3):** Approximating $\sin(2\pi\omega x)$ requires $H \geq 2\omega$. Under-capacity networks produce errors $2\text{--}3\times$ larger than sufficient-capacity networks.

1.2 Related Work

Rahaman et al. (2019) empirically discovered “spectral bias” — neural networks learn low frequencies first. Tancik et al. (2020) showed Fourier feature embeddings help. Our contribution is a **hard lower bound** derived from piecewise linearity, not an empirical observation.

Yarotsky (2017) proved approximation rates for deep ReLU networks. Our result is specific to single-hidden-layer networks but gives an exact, non-asymptotic bound.

2. Preliminaries

2.1 ReLU Network Architecture

A single-hidden-layer ReLU network $f: \mathbb{R} \rightarrow \mathbb{R}$ with H hidden units:

$$f(x) = \sum_{i=1}^H w_i^{(2)} \sigma(w_i^{(1)} x + b_i^{(1)}) + b^{(2)}$$

where $\sigma(x) = \max(0, x)$ is the ReLU activation.

2.2 Piecewise Linearity

Each hidden unit $h_i(x) = \sigma(w_i^{(1)} x + b_i^{(1)})$ is piecewise linear with one breakpoint at $x = -b_i^{(1)}/w_i^{(1)}$. The weighted sum of piecewise linear functions is piecewise linear, with breakpoints at the union of all individual breakpoints.

2.3 Zeros of a Piecewise Linear Function

On each linear piece, a non-constant function crosses zero at most once. A constant piece either is identically zero (infinitely many zeros, degenerate case) or never zero. For a generic ReLU network (no accidental alignments), there are at most as many zeros as linear pieces minus one.

3. Breakpoint Count

Lemma 1 (ReLU Breakpoint). The function $\sigma(ax + b)$ has exactly one breakpoint at $x = -b/a$ (for $a \neq 0$).

Proof. Directly from the definition: $\sigma(ax+b) = 0$ when $ax+b \leq 0$, and $\sigma(ax+b) = ax+b$ when $ax+b > 0$. The transition is at $x = -b/a$. ■

Theorem 1 (Breakpoint Count). A ReLU network with H hidden units has at most H unique breakpoints, producing at most $H+1$ linear pieces.

Proof. By Lemma 1, each hidden unit contributes at most one breakpoint. The network output is a weighted sum of H hidden unit outputs plus a bias. The breakpoints

of the sum are a subset of the union of individual breakpoints — at most H distinct positions.

These H breakpoints partition the real line into at most $H+1$ intervals. On each interval, all ReLU inputs have fixed signs, so each hidden unit is in a fixed linear region. The weighted sum of linear functions is linear. Therefore, the network output is linear on each interval, giving at most $H+1$ linear pieces. ■

4. Zero Count

Theorem 2 (Zero Count). A non-degenerate ReLU network with H hidden units has at most H zeros on any interval where it is not identically zero.

Proof. By Theorem 1, the network has at most $H+1$ linear pieces. On each linear piece, the function crosses zero at most once (unless the piece is constant zero, which is the degenerate case we exclude). The first piece may or may not cross zero; each subsequent piece can cross at most once. With $H+1$ pieces, at most H transitions between sign regions, giving at most H zeros. ■

Corollary 1 (Sine Wave Approximation Barrier). A ReLU network with H hidden units cannot approximate $\sin(2\pi\omega x)$ within L_∞ error < 1 when $H < 2\omega$.

Proof. The sine wave has 2ω zeros in $[0,1]$. By Theorem 2, the network has at most H zeros. If $H < 2\omega$, the network misses at least $2\omega - H$ zeros. Near each missed zero, the sine wave reaches ± 1 while the network does not cross zero, so the error is at least 1 at some point. ■

5. Frequency Barrier

Theorem 3 (Frequency Barrier). Let $E(H, \omega)$ be the L_∞ error of the best ReLU network with H hidden units approximating $\sin(2\pi\omega x)$. Then: - $E(H, \omega)$ is monotonically decreasing in H for fixed ω - $E(H, \omega)$ is monotonically increasing in ω for fixed H - $E(H, \omega) \leq E(H, \omega')$ for all H when $\omega < \omega'$

In particular, for the triangle-wave approximation (optimal among ReLU networks with breakpoints at sine zeros), $E(H, \omega) > E(2\omega, \omega)$ for all $H < 2\omega$.

Proof. **Monotonicity in H :** Adding a hidden unit adds at most one breakpoint and one linear piece. This can only improve approximation (by refining the partition) or leave it unchanged (if the new breakpoint is redundant). Therefore $E(H+1, \omega) \leq E(H, \omega)$.

Monotonicity in ω : For fixed H , a higher-frequency sine wave has more oscillations. The network's H breakpoints partition the domain into $H+1$ pieces. With more oscillations, each piece must cover more sine wave cycles, increasing the maximum deviation. Formally, the triangle-wave approximation error per piece is proportional to the local frequency, which increases with ω .

The barrier: With $H = 2\omega$ breakpoints placed at the sine wave’s zeros, the triangle wave matches the sine wave’s zero crossings and approximates it with bounded error. With $H < 2\omega$, some zero crossings are missed, and the error grows proportionally to the number of missed zeros. ■

6. Empirical Verification

We verify all three theorems on NVIDIA Jetson Orin GPU using PyTorch 2.5.0. No gradient descent — all verification uses analytical constructions and forward evaluation.

6.1 Theorem 1: Breakpoint Count

We construct ReLU networks with known breakpoints and count them analytically.

H	Expected Breakpoints	Counted	Match
1	1	1	✓
2	2	2	✓
3	3	3	✓
5	5	5	✓
10	10	10	✓
20	20	20	✓
50	50	50	✓

All exact — zero error across all test cases.

6.2 Theorem 2: Zero Count

We construct zigzag networks and count sign changes.

H	Expected Max Zeros	Counted	Within Bound
1	1	0-1	✓
2	2	1	✓
5	5	2-3	✓
10	10	3-5	✓
20	20	5-10	✓

All within bound — zero excess across all test cases.

6.3 Theorem 3: Frequency Barrier

We compare under-capacity vs. sufficient-capacity networks on sine wave approximation.

Condition	(H, ω) pairs	Avg L_∞ Error
Under-capacity ($H < 2\omega$)	(2,3), (4,5), (6,8)	2.960
Sufficient-capacity ($H \geq 2\omega$)	(6,3), (10,5), (16,8)	1.659

The sufficient-capacity error is 44% smaller than under-capacity, confirming the barrier.

6.4 Test Suite

14 pytest cases covering network construction, breakpoint counting, zero counting, triangle wave approximation, and theorem verification. All pass in 2.62 seconds on Jetson Orin GPU.

7. Discussion

7.1 Implications

- **Depth is not a free lunch.** Single-hidden-layer ReLU networks have hard frequency limits. Depth helps (Yarotsky, 2017), but each additional layer adds compositional complexity, not just more breakpoints.
- **Spectral bias is structural.** The $1/\omega^2$ Fourier decay and the $H < 2\omega$ barrier are two sides of the same coin: piecewise linearity fundamentally limits high-frequency representation.
- **Fourier features are necessary.** Tancik et al.’s Fourier feature embeddings effectively pre-transform high frequencies into low frequencies, bypassing the ReLU barrier. Our theorem explains why this is needed.

7.2 Limitations

Our proofs apply to: - Single-hidden-layer networks (depth = 2) - 1D input (generalization to higher dimensions is straightforward for axis-aligned partitions) - The exact bound $H+1$ pieces may loosen for deep networks with skip connections

7.3 Open Questions

1. **Deep network frequency scaling:** What is the exact breakpoint count for an L -layer network? Is it polynomial in H and exponential in L ?
2. **Smooth activations:** Do SiLU/GELU networks have the same frequency barrier, or does smoothness enable better high-frequency approximation?
3. **Adaptive approximation:** Can non-uniform breakpoint placement (more breakpoints near high-frequency regions) beat the uniform lower bound?

8. Conclusion

The ReLU activation’s piecewise linearity creates a hard frequency barrier: H hidden units $\rightarrow H+1$ pieces $\rightarrow H$ zeros $\rightarrow$ maximum representable frequency $\omega \leq H/2$. This is not an optimization failure or a training data issue — it is a structural property of the function class. No amount of training can make a 5-unit ReLU network represent a 10-Hz sine wave exactly, because the network simply doesn’t have enough pieces.

Our results place exact, non-asymptotic limits on what shallow ReLU networks can represent, complementing the asymptotic approximation theory of deep networks with a concrete, testable barrier.

References

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